

Homework #5 Lexi, Command, ChainOfResponsibility, and Prototype

Issued: Monday, April 13

Due: Monday, April 27

Purpose

This assignment allows you to learn about the Command(233), ChainOfResponsibility(223), and Prototype(117) design patterns. You should also use Singleton(127).

Assignment

Design (in UML) and implement (in Java) the User Operations part of the Lexi editor, as described in Section 2.7 of our textbook.

Your program should accept the following keystrokes and perform the following operations:

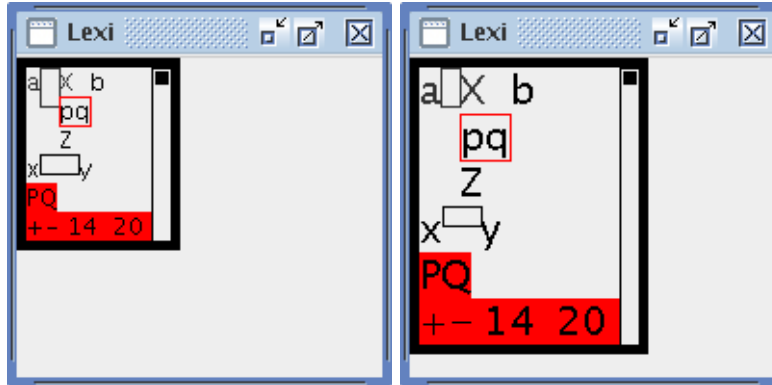
- i: Increment the font size by one.
- d: Decrement the font size by one.
- u: Undo.
- r: Redo.

Your “main program” should construct and display buttons, which when clicked perform the following operations:

- Increment the font size by one.
- Decrement the font size by one.
- Set the font size to fourteen.

- Set the font size to twenty.

Test your solution with a simple graphical demonstration. For example, these are the results of my tests:



Notes and Suggestions

- Use the design suggestions from our textbook.
- Use the Java code in:
 pub/hw5
- Java's Swing and AWT window systems recognize keystrokes and mouse clicks. Both call `keyTyped()` and `mouseClicked()` methods on `PaneKeyListener` and `PaneMouseListener` objects, respectively. These, in turn, call `key()` and `click()` methods on your `ApplicationWindow` object, from the previous assignment. So, add `key()` and `click()` methods in your `Window` class.
- As usual, casting is prohibited!
- As usual, indicate with a comment at the top of each source file, the relevant patterns and participants.